

Network Forensic Report

PCAP Network Packet Capture Analysis

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1. Executive Summary

The goal of this assessment centers on investigating the contents of network capture data file "XYZ.pcap" documenting network activities that occurred within an educational institution. The analysis team extracted the file to conduct its assessment directly on an internal hard drive. The investigation worked towards rebuilding the network infrastructure alongside critical device identification while understanding the activities that occurred before and during a harassment complaint concerning Professor Lily Tuckrige. Throughout the examination process the investigators used Wireshark as the main network forensic tool.

1. The examination centered on subnet 192.168.15.xx with identification of 17 active devices. The main device implicated in the investigation used Apple hardware operating under MAC address 00:17:f2:e2:c0:ce with corresponding IP address 192.168.15.4.
2. The suspect's device established connection with an external IP address 140.247.62.34 and the domains sendanonymousemail.net and Willselfdestruct.com.
3. Professor Tuckrige received harassing messages at 140.247.62.34 which is the public IP address of the device HonHaiPrecis_2e:4f:60 which we identified as a router.
4. Further look into the analysis through packet capture filters made us find the suspect who uses the Gmail jcoachj@gmail.com.
5. The investigation discovered the suspect searching for ways to harass teachers and making attempts to use anonymous services online.
6. Enrollment records along with activity log evidence identified the suspect who was in Chemistry Class 109.
7. Multiple pieces of evidence show the suspect hid their identity through virtual machines along with anonymizing platforms.
8. The analysis of suspect's computer activities uncovered his access even though the website operated as a previous source of harassment against the target.

The document outlines in depth detail how the network functions and the technologies used during the investigation to establish their findings about cyber harassment cases.

2. Introduction

2.1 Network Capture File details

The forensic parameters of the extracted PCAP file XYZ.pcap is given below. The evidence for these details is provided in Figure 1 extracted from Wireshark ver4.4.3:

Capture length:	56MB
Format:	Wireshark/tcpdump/... -pcap
Packet size limit:	4096 bytes
First packet:	22-JULY-2008 07:21:07 HRS
Last packet:	22-JULY-2008 11:43:47 HRS
Elapsed time:	4 hours, 22 minutes, 39 seconds (15759.951 seconds)
Total packets:	94,410
Average packets/sec	5.991 packets/sec
Average packet size	579 bytes
Average bytes/sec	3468 bytes/sec

Computed HASHes – xyz.pcap

Hash (SHA256) :

2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb

Hash (SHA1) :

65656392412add15f93f8585197a8998aaeb50a1



Figure 1. Packet capture summary from Wireshark ver4.4.3

2.2 Network Components Identified

There are 17 different ethernet components that can be identified using the Ethernet Endpoints listed under Statistics as below. The below evidences are provided by Wireshark ver4.4.3.

Address	Packets	Bytes	Tx Packets	Tx Bytes	Rx Packets	Rx Bytes
IPv4mcast_09	1	70 bytes	0	0 bytes	1	70 bytes
IPv4mcast_16	3	192 bytes	0	0 bytes	3	192 bytes
Apple_e2:c0:cf	8	512 bytes	0	0 bytes	8	512 bytes
IPv4mcast_02	9	540 bytes	0	0 bytes	9	540 bytes
Apple_f1:8a:6e	11	856 bytes	7	474 bytes	4	382 bytes
Apple_b4:a3:f8	25	2 kB	25	2 kB	0	0 bytes
IPv4mcast_fb	33	2 kB	0	0 bytes	33	2 kB
Apple_79:00:31	63	4 kB	63	4 kB	0	0 bytes
TRENDnet_44:a0:f1	74	6 kB	74	6 kB	0	0 bytes
Apple_69:38:cc	75	3 kB	73	3 kB	2	128 bytes
Apple_5a:77:9b	187	17 kB	176	11kB	11	6 kB
IPv4mcast_7f:ff:fa	2225	819 kB	0	0 bytes	2225	819 kB
Broadcast	3941	252 kB	0	0 bytes	3941	252 kB
HonHaiPrecis_2e:4f:61	14990	9 MB	6916	1MB	8074	8 MB
Commscope_99:98:68	18499	9 MB	11633	8 MB	6866	1 MB
Apple_e2:c0:ce	73246	45 MB	34582	6 MB	38664	39 MB
HonHaiPrecis_2e:4f:60	75430	46 MB	40861	40 MB	34569	6 MB

Table 1. Endpoints lists from Wireshark ver4.4.3

The device HonHaiPrecis_2e:4f:60 whose IP address is 192.168.15.1 has had communication with all devices in the network. It has the highest number of packets (75430) and bytes (46 MB). It's been sending and receiving lots of traffic including TCP packets. This device has transported requests from the suspect's IP to outer source. Also there are several IP addresses for its MAC address 00:1d:d9:2e:4f:60 which means it acts as the router of the network. There are several Apple devices. Out of those, the device Apple_e2:c0:ce whose IP address is 192.168.15.4 has large amount of communication. It has 73246 packets. The Figure 2 shows the MAC addresses of each device mentioned in the Table 1 above.

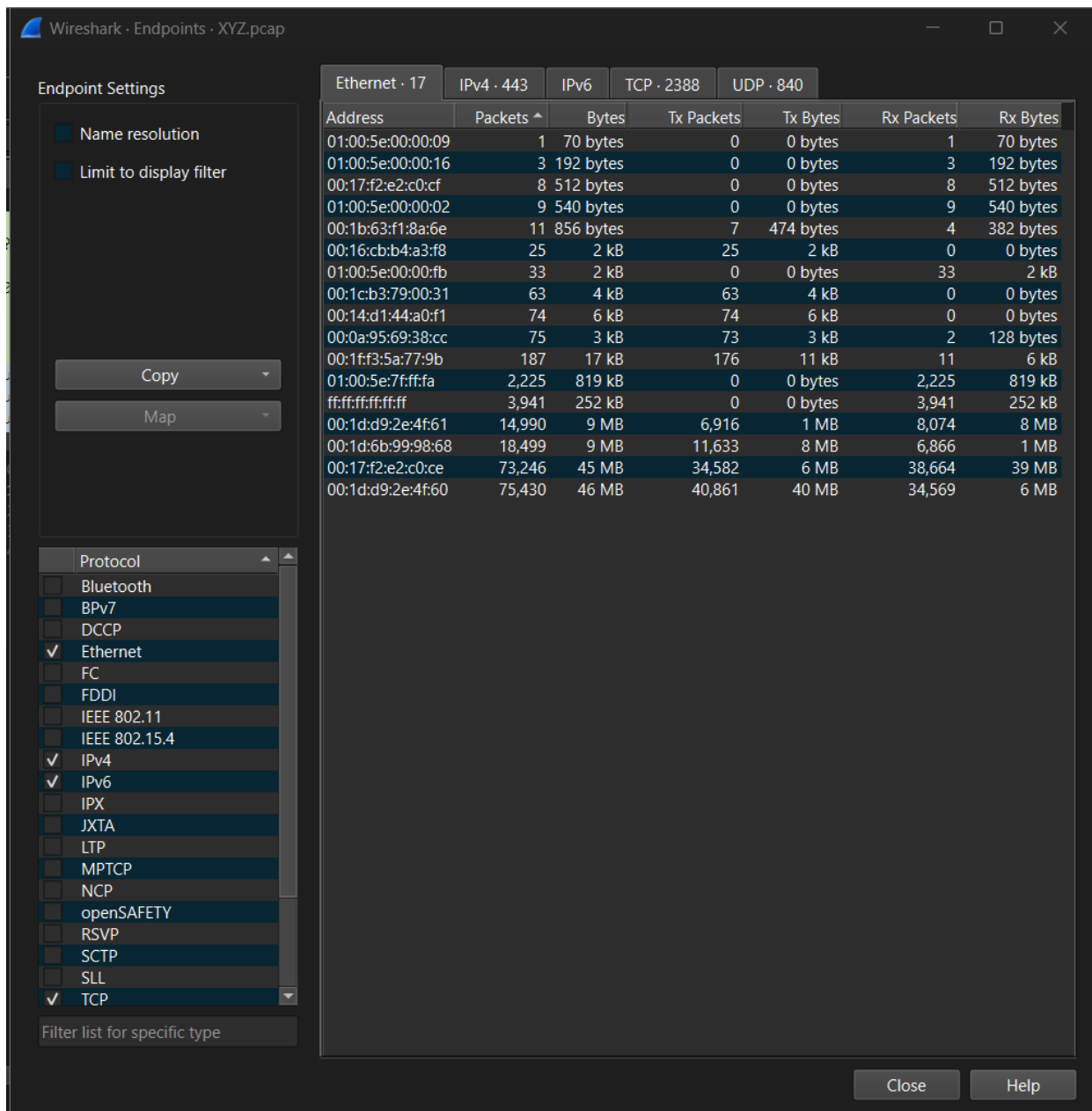


Figure 2. Endpoints from Wireshark ver4.4.3

3. Methodology

3.1 Tools Used

Implementation of Wireshark Version 4.4.3 performed a wide examination specifically to find background evidences on the PCAP file xyz.pcap which took place on a computer operating Windows 11 Home 64-bit platform and an Intel(R) Core(TM) i3 processor. The PC operates with a base frequency measuring 3.00 GHz and possesses 8 GB of Random Access Memory (RAM). Wide popularity of Wireshark made it a suitable choice for thoroughly investigating the network traffic recorded in the PCAP file.

For the cross examination including the background evidence findings were analysed independently within a machine equipped with an operating system Windows 10 Pro 64 bit using equipment that consisted of an Intel(R) Core(TM) i3 processor and 12GB of RAM.

3.2 Steps Involved

The downloaded XYZ.pcap file was opened using Wireshark v4.3.3 in Windows for the analysis. The statistic tab provides list of endpoints and conversations on Ethernet, IP, TCP and UDP layers. The following filters were used to analyse the pcap file based on the IPs that were most crucial to this incident. By doing so, it made it easier for us to analyse network behaviour and detect potential issues.

TCP

When used in Wireshark, this filter captures only TCP traffic by isolating TCP-based connections by displaying the packets that use only TCP protocol. We used this filter because, TCP is the most common protocol used for reliable communication on the internet like for web browsing and file transfer.

HTTP

This filter captures the packets that only use HTTP protocol which means only web traffic between clients and web browsers which made it easier for us to investigate web browsing activity, to identify the websites users visit and to find any suspicious searches done by the user.

DNS

This filter captures the packets that shows all DNS lookups happening on the network by which we can identify which websites are being resolved or accessed. This is useful when looking for suspicious domain names or phishing sites.

ip.addr == 192.168.15.4

This filter captures all traffic when Ip 192.168.15.1 is either source or destination IP address. So, we can track the specific device having this particular IP address and also, we can monitor all incoming and outgoing network traffic of that device.

http && ip.addr == 192.168.15.4

This combined filter captures all packets that only use HTTP protocols whose source or destination IP address is 192.168.15.4 by which we can analyse web traffic associated with a particular device.

ip.addr == 192.168.1.254

This filter captures all traffic when Ip 192.168.1.254 is either source or destination IP address by which we can track the specific device having this particular IP address and also we can monitor all incoming and outgoing network traffic of that device.

dns && ip.addr == 192.168.1.254

This filter captures all packets that contain only DNS communication to and from the IP address 192.168.1.254 by which we can monitor the DNS traffic associated with a particular device on the network.

ip.dst == 140.247.62.34

This filter captures all the packets where destination IP address is 140.247.62.34 by which we can monitor incoming traffic to a specific IP address.

ip.src == 140.247.62.34 or ip.dst == 140.247.62.34

This filter captures all the packets where either source or destination IP address is 140.247.62.34 by which we can monitor incoming traffic to and from a specific IP address.

frame contains "send+anonymous"

This filter captures the packets which contain the text "send+anonymous" which is helpful in identifying suspicious data, malicious activity or email messages.

frame contains "tuckrige"

This filter captures the packets which contain the text "tuckrige" which is helpful in analysing packets containing suspicious messages which are related to a specific person

.

ip.addr == 192.168.15.4 and frame contains "mail"

This filter captures the packets containing the text "mail" which is helpful in looking for email related traffic from or to a suspicious IP address. This filter is useful in situations where we need to find suspicious email communications, harassing emails and emails sent via non-standard channels.

YMSG

This filter isolates only Yahoo Messenger traffic from other packets which is helpful in analysing instant messaging activity on a Yahoo network if we want to look for suspicious behaviour in communications.

3.3 Handling Data

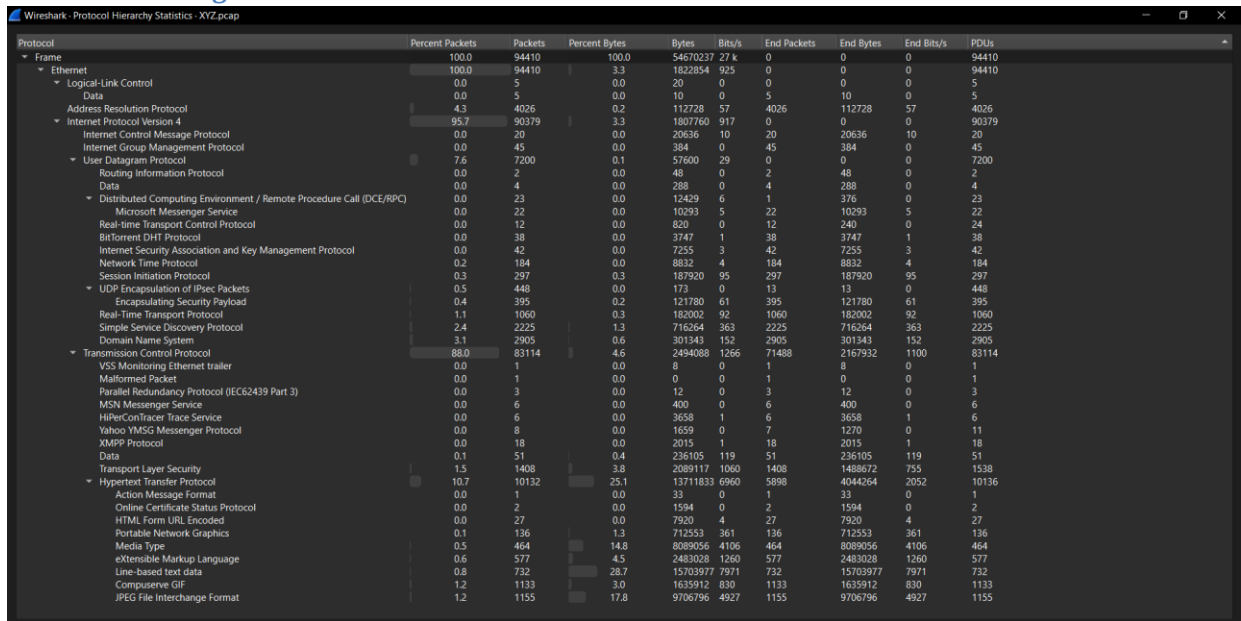


Figure 3. Protocol Hierarchy captured with Wireshark ver4.4.3

Packet Type	Number of packets	Requests	Technique	Responses	Technique
ARP	4026	3972	arp.opcode == 1	54	arp.opcode == 2
IP	90379	-	-	-	-
TCP	83114	2169	tcp.flags.syn == 1 and tcp.flags.ack == 0	1974	tcp.flags.syn == 1 and tcp.flags.ack == 1
UDP	7200	-	-	-	-
DNS	2905	-	-	-	-
HTTP	10132	-	-	-	-
SSL	1408	-	-	-	-
MSNMS	6	-	-	-	-

Table 2. Decomposition of different packet types from capture

Based on the statistics given in Table 2, the network was predominantly TCP traffic based. The Wireshark tcp filters reveal significant amount of TCP request responses to IP addresses 192.168.15.4 and 192.168.1.64. Since the machine with IP 192.168.15.1 appears to be a Router, this report concludes that 192.168.15.4 was the major contributor to this traffic.

IP Address	Mac Address	As Sender	As Receiver
140.247.62.34	HonHaiPrecis_2e:4f:60	52	65
192.168.15.4	Apple_e2:c0:ce	34,554	38,643
192.168.15.1	HonHaiPrecis_2e:4f:60	2,154	-
192.168.15.2	Apple_f1:8a:6e	2	4
192.168.1.64	Commscope_99:98:68	6,814	8,084

Table 3. Decomposition of IP traffic (Inbound & Outbound)

The information gathered from the table is enough to conclude that user 192.168.15.4 is the single most traffic generating user on this network within the time the network usage was being recorded.

4. Detailed Findings

4.1 Important network players

Host Name	MAC address	IP address	Device/ Owner
HonHaiPrecis_2e:4f:60	00:1d:d9:2e:4f:60	192.168.15.1	Router
Apple_e2:c0:ce	00:17:f2:e2:c0:ce	192.168.15.4	Johny Coach

Table 4. Important Network Players

Based on the analysis, we identified that there is a crucial link between the IP addresses 192.168.15.1 and 192.168.15.4. The device HonHaiPrecis_2e:4f:60 has exhibited multiple IP addresses linked to external services like Amazon, Google, YouTube and Yahoo which made us identify that this device acts as a router or a gateway interacting with devices on the inside and outside of the network. Addition to this, we looked for source and destination addresses on ethernet and IP packets and it led us to the IP address 192.168.15.4 which was constantly present in HTTP requests. This is the Apple device corresponding to the MAC address 00:17:f2:e2:c0:ce that we earlier identified as having lots of traffic by which we discovered that this device has involved in suspicious activity.

4.2 Network Structure

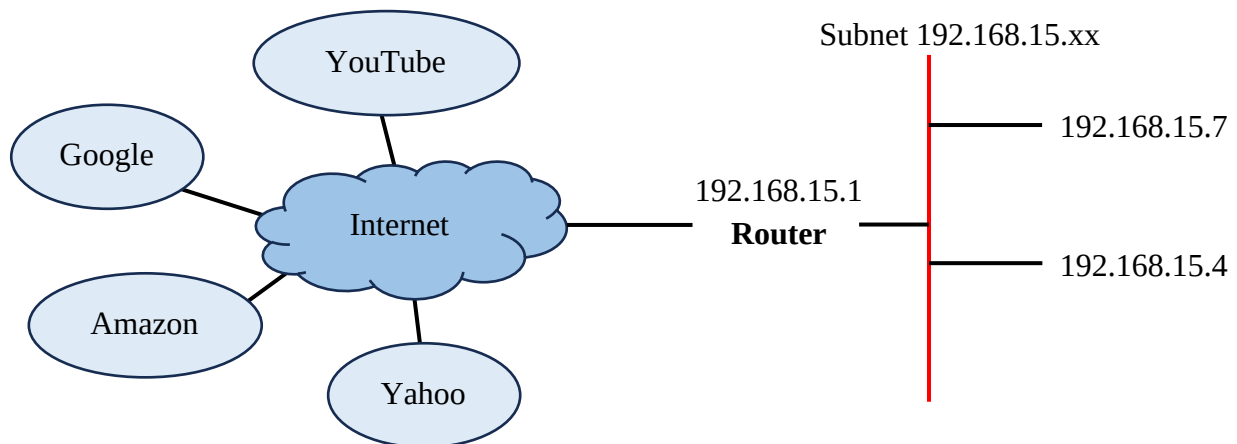


Figure 4. Possible Network Structure based on reconstruction from XYZ.pcap

The above image shows the network structure we assumed after the further analysis of network traffic. With the filter `http && ip.addr == 192.168.15.4`, we could identify that the suspicious device has sent HTTP requests over the network to the device HonHaiPrecis_2e:4f:60 which were to access various suspicious websites related to the incident. When we examine the packets, we found that there are several DNS requests to the device under the IP address 192.168.1.254 whose MAC address is 00:1d:d9:2e:4f:60 from the suspicious IP 192.168.15.4. So, we believe that the router HonHaiPrecis_2e:4f:60 acts as a DNS server with the IP 192.168.1.254

4.3 Activity Timeline for 192.168.15.4

Packet Number	Activity	Destination	Inference
25080	Referer: http://kids.yahoo.com	209.73.191.242	Search Yahoo for Kids Parents Blog article discussing whether the movie <i>The Dark Knight</i> (2008) is suitable for children.
26258	Referer: http://www.google.com/search?client=safari&rls=en-us&q=dark+knight+trailer&ie=UTF-8&oe=UTF-8\r\n	74.125.19.99	Searched for <i>The Dark Knight</i> (2008) movie trailer on Google using Safari as their web browser.
27472	Referer: http://www.youtube.com/watch?v=WaIR9dAZRR0\r\n	208.65.153.251	This likely refers to a video related to <i>The Dark Knight</i> (2008), possibly a trailer or a scene from the movie .
35092	Referer: http://www.amazon.com/gp/product	216.38.162.97	Search for a product on amazon.

Table 5. Activity Timeline based on network forensic analysis and event reconstruction

Note: Only distinct IP addresses with different time stamps are mentioned on the table. Each IP has a sequence of following TCP/HTTP packets following it which is not captured in this table. Packet numbers provided in the table indicate the first occurrence of the transaction.

4.4 Background evidence

1. The capture file is confirmed to be forensically secure by the MD5SUM of the decompressed file compared with the report provided. Furthermore, additionally secured this copy by the SHA1SUM and the SHA256SUM of the same file. See Section 2.1 for further details.
2. Between July 22, 2008, 01:51:07.095278 UTC to July 22, 2008, 06:13:47.046029 UTC, the network traffic under examination took place. This period of time acts as the main focus for examining the information and incidents linked to the suspect's actions.
3. We discovered 17 devices that were accessible within the network when we looked at its endpoints.
4. The subnet is 192.168.15.XX and the assigned IP address 192.168.15.1 owns responsibility for gateways and hub control functions in network communication. Detailed exploration of this IP address would clarify its significance for network design. The core aspects of the network can be identified through taking a deeper look at network relations and operations. The main investigative element comprises the suspect's IP address at 192.168.15.4
5. This **ip.addr == 192.168.15.4** filter enables users to monitor all network events tied to IP address 192.168.15.4. The analyzed IP stands as a private IP which represents the public address assigned to the router that connects internal systems to outside networks. The diagnostic process traced network communications starting from or ending at 192.168.15.4 which discovered that the device engaged in sending harassing emails to lilytuckrige@yahoo.com through sendanonymousemail.net and Willselfdestruct.com services. The MAC address 00:17:f2:e2:c0:ce belonging to 192.168.15.4 led to identification of an Apple device. Network experts used this specific information to identify which device among all others was causing harmful activity in the system.
6. By applying the filter **ip.src == 140.247.62.34 or ip.dst == 140.247.62.34**, we can identify all network traffic where the IP address 140.247.62.34 is either the source or destination. The filter data shows 192.168.15.4 as the single IP address that transmits and receives traffic from 140.247.62.34. The findings demonstrate that 192.168.15.4 connects the internal network to the external IP address 140.247.62.34. The analysis reveals that 192.168.15.4 belongs to Apple devices.
7. The filter **frame contains "tuckrige"** shows three essential packets which relate to the harassment investigation of Professor Lily Tuckrige. The first two packets serve the application layer via HTTP service whereas the third operates through the transport layer using TCP. An anonymous harassing message from IP address 192.168.15.4 that originated on sendanonymousemail.net was dispatched toward 69.80.225.91 according to packet 80614. The excessive message transmitted via Willselfdestruct.com can be found in packet number 83601 where the email header contains the phrase "you can't find us." The message originated from source IP address 192.168.15.4 as recorded in the packet while going to destination IP address 69.25.94.22. Research shows how the IP address 192.168.15.4 plays an essential part in distributing the harassing messages sent to Professor Tuckrige.

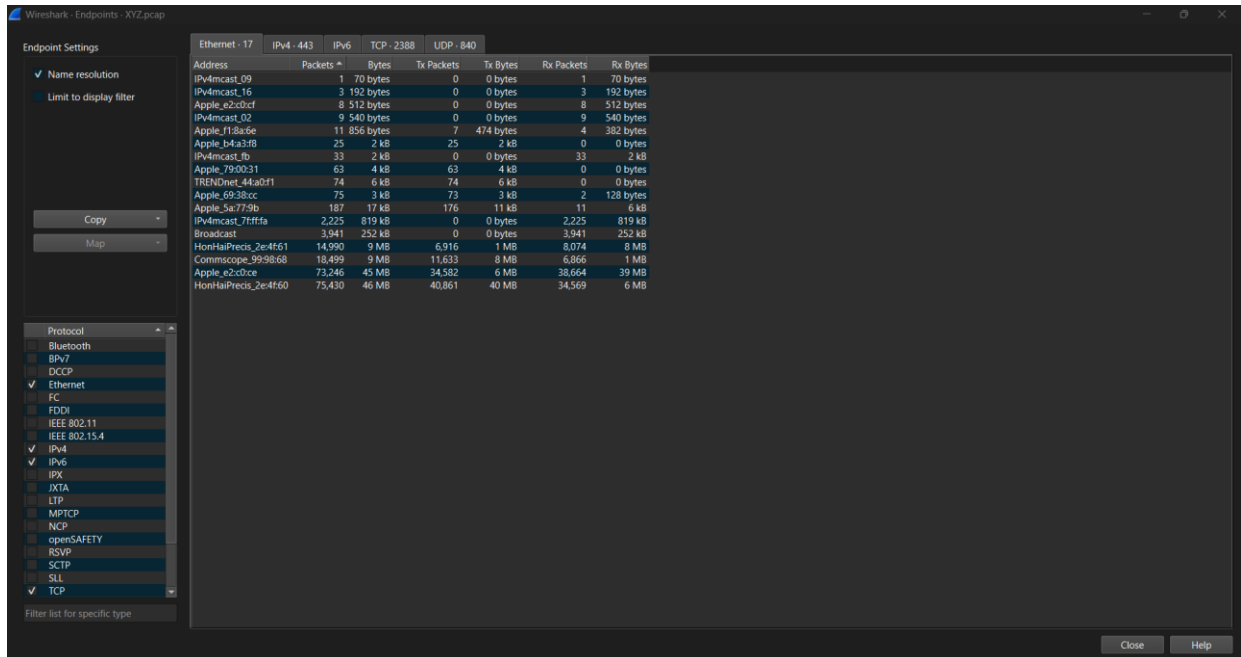
8. The packets were analyzed through **ip.addr == 192.168.15.4** and frame contains "mail" filtering to identify the attacker's email address. The email message at packet number 77528 shows an address of jcoachj@gmail.com.
9. Investigation of Johnny Coach's search activities revealed his desire to send anonymous emails through the internet. He started his research by entering "send anonymous mail" into Google which directed him to additional websites he visited. His visit to www.sendanonymousemail.net demonstrated his interest in the practical usage of services for electronic contact with hidden sender identification. The information release about such anonymizing services generated worries regarding their future misuse. During the ongoing study Johnny became focused on reading About.com's explanation about the problems in sending anonymous emails. Johnny made a specific effort to obtain information and resources for maintaining anonymous communication on the internet. Evidence determined that Johnny visited a page linked to sending the latest threatening message forcing the situation to become even more critical. The URL (http://clk.about.com/?zi=18/1R4/Wa&sdn=email&cdn=compute&tm=10&gps=317_269_788_511&f=00&tt=14&bt=1&bts=1&zu=http%3A//email.about.com/od/anonymousemailservices/gr/will_self_destr.htm) suggested that he utilized a particular platform to transmit risky or threatening messages. The discovered evidence of Johnny Coach's digital actions leads to grave concerns about his actions while requiring extensive investigation to determine both the extent of his online activity and potential legal consequences.
10. A crucial evidence discovery from the investigation indicates clear intentions of the suspect. A search made by the suspect was uncovered by thorough network traffic analysis. The user from IP address 192.168.15.4 performed online searches with disturbing intentions which included "want to harass my teacher." Knowledge gained from this specific search request helps experts determine the suspect's motivation and reveals their possible connection with both dangerous actions and illegal conduct. The search was uncovered by applying the filter **http && ip.addr == 192.168.15.4** that permitted investigators to study HTTP activities from the suspect. The investigation revealed the suspect directly searched for teacher harassment information through their activity during the observation period. The search query establishes alarming intentions about the suspect while serving as a starting point to investigate online behaviour patterns. Further analysis of supplemental search history and websites and related behavioural contexts will help create a thorough picture of the suspect's digital tracks. The investigator can determine the suspect's intentions because network logs have recorded their search inquiry. The suspect performed a search which exposed his intentions to bother his teacher. A search query presents important evidence revealing both the suspect's intentions along with their potential unlawful actions.
11. Our investigation involved a supplementary step which compared digital evidence against factual information. The analysis of network activities linked to IP address 192.168.15.4 revealed the presence of the user Johnny Coach. Our investigation of this user identity required us to consult both local context and the student enrolment list of chemistry class 109. The research produced an important discovery about Johnny Coach being an enrolled student in the chemistry class. The link between digital evidence and human identities brings our study into contact with the real person involved in the activities.

12. Evidence from the investigation revealed that Johnny Coach gained access to nitroba.org which hijacked investigative attention about his online activities. The discovery of this domain holds critical importance because nitroba.org was previously detected as the origin that sent harassment emails to Miss Lilly Tuckerige. The domain nitroba.org represents the first sender of email harassment to Miss Lilly Tuckerige. The relationship between Johnny Coach and nitroba.org domain creates important investigative questions about his potential role in organizing or helping the harassing messages.
13. The investigation has experienced major developments because investigators found the suspect's identity. Analyses of the logs established that IP address 192.168.15.4 belongs to Johnny Coach as per our investigation logs. The analysis of HTTP history revealed revelation of information including the Gmail account. The system recorded entry from jcoachj@gmail.com which investigators matched to the name Johnny Coach. Log number 77528 contains the cookies which reveal the identity.
14. In our investigation we used the filter **frame contains "tuckrige"** to find packets of interest during the analysis of captured traffic. The MAC address Apple_e2:c0:ce (00:17:f2:e2:c0:ce) exists within the filtered packets thus indicating the physical hardware connects to an Apple device. The same packet disclosed this User-Agent string: Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1; SV1)\r\n. Windows XP operating system type appears within the User-Agent string along with identification of Windows NT 5.1. The inconsistent details indicate that the person ran Windows XP operating system through a virtual machine that operated on Apple products. A VM usage is apparent because the reported operating system information in the User-Agent string (Windows NT 5.1) does not match the actual hardware (Apple).
15. The logs analysed with the **YMSG** filter revealed a vital entry that referenced amy789smith as an active user. The momentous discovery motivated us to follow the digital trail intensively in order to understand "amy789smith's" involvement better. Our research into the class's student population confirmed "amy789smith" as one of its members. Our research to tie the student to reported teacher harassment revealed no evidence showing the student took part in unprofessional conduct. The truth-seeking nature of our research revealed that Amy Smith as a student has not engaged in the alleged harassing activities.

5. Supporting Evidence Presented

Evidence 01

Network evidence: Ethernet Endpoints



Wireshark - Endpoints - XYZ.pcap

Endpoint Settings

- ✓ Name resolution
- Limit to display filter

Copy

Map

Protocol

- Bluetooth
- BPv7
- DCCP
- ✓ Ethernet
- FC
- FDDI
- IEEE 802.11
- IEEE 802.15.4
- ✓ IPv4
- ✓ IPv6
- IPX
- JXTA
- LTP
- MPTCP
- NCP
- openSAFETY
- RSVP
- SCTP
- SLL
- ✓ TCP

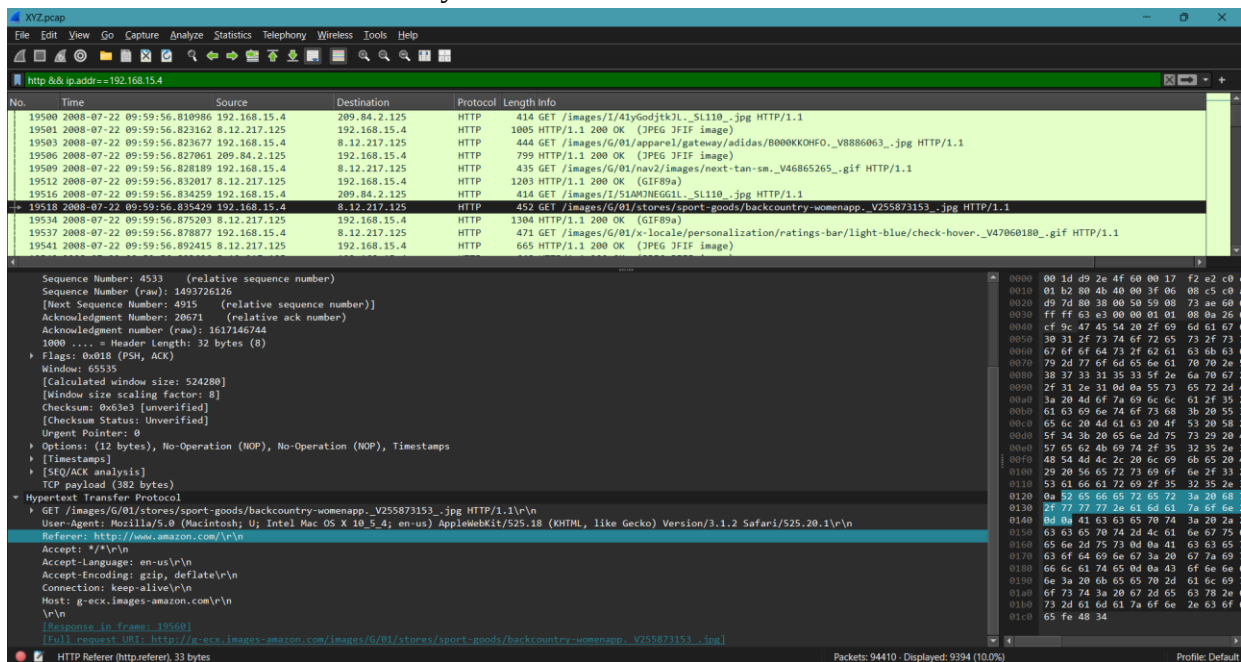
Filter list for specific type

Address	Packets	Bytes	Tx Packets	Tx Bytes	Rx Packets	Rx Bytes
IPv4mcast_09	1	70 bytes	0	0 bytes	1	70 bytes
IPv4mcast_16	3	192 bytes	0	0 bytes	3	192 bytes
Apple_e2cdcf	8	512 bytes	0	0 bytes	8	512 bytes
IPv4mcast_02	9	540 bytes	0	0 bytes	9	540 bytes
Apple_f18a6e	11	856 bytes	7	474 bytes	4	382 bytes
Apple_14a318b	25	2 kB	25	2 kB	0	0 bytes
IPv4mcast_1b	33	2 kB	0	0 bytes	33	2 kB
Apple_790031	63	4 kB	63	4 kB	0	0 bytes
TRENDnet_44a0f1	74	6 kB	74	6 kB	0	0 bytes
Apple_6938cc	75	3 kB	73	3 kB	2	128 bytes
Apple_5a779b	187	17 kB	176	11 kB	11	6 kB
IPv4mcast_7ffffa	2,225	819 kB	0	0 bytes	2,225	819 kB
Broadcast	3,941	252 kB	0	0 bytes	3,941	252 kB
HonHaiPrecis_2e4f61	14,990	9 MB	6,916	1 MB	8,074	8 MB
Commscope_999868	18,499	9 MB	11,633	8 MB	6,866	1 MB
Apple_e2cdce	73,246	45 MB	34,582	6 MB	38,664	39 MB
HonHaiPrecis_2e4f60	75,430	46 MB	40,861	40 MB	34,569	6 MB

Close Help

Evidence 02

Evidence of searches conducted by the user



XYZ.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

http && ip.addr==192.168.15.4

No.	Time	Source	Destination	Protocol	Length	Info
19508	2008-07-22 09:59:56.810986	192.168.15.4	209.84.2.125	HTTP	414	GET /images/i/4lygodjtk3L_S1130_.jpg HTTP/1.1
19501	2008-07-22 09:59:56.823162	8.12.217.125	192.168.15.4	HTTP	1005	HTTP/1.1 200 OK (JPEG 7F1F image)
19503	2008-07-22 09:59:56.823677	192.168.15.4	8.12.217.125	HTTP	444	GET /images/G/01/apparel/gateway/adidas/B00NKKHFO_V8886063_.jpg HTTP/1.1
19506	2008-07-22 09:59:56.827061	209.84.2.125	192.168.15.4	HTTP	799	HTTP/1.1 200 OK (JPEG 7F1F image)
19509	2008-07-22 09:59:56.828189	192.168.15.4	8.12.217.125	HTTP	435	GET /images/G/01/nav2/images/next-tan-sm_V46865265_.gif HTTP/1.1
19512	2008-07-22 09:59:56.832017	8.12.217.125	192.168.15.4	HTTP	1203	HTTP/1.1 200 OK (GIF89a)
19515	2008-07-22 09:59:56.834259	192.168.15.4	209.84.2.125	HTTP	418	GET /images/I/51AMNNGGIL_S1130_.jpg HTTP/1.1
19518	2008-07-22 09:59:56.835429	192.168.15.4	8.12.217.125	HTTP	452	GET /images/G/01/stores/sport-goods/backcountry-womenapp_V255873153_.jpg HTTP/1.1
19534	2008-07-22 09:59:56.875203	8.12.217.125	192.168.15.4	HTTP	1304	HTTP/1.1 200 OK (GIF89a)
19537	2008-07-22 09:59:56.878877	192.168.15.4	8.12.217.125	HTTP	471	GET /images/G/01/x-locale/personalization/ratings-bar/light-blue/check-hover_V47060180_.gif HTTP/1.1
19541	2008-07-22 09:59:56.892415	8.12.217.125	192.168.15.4	HTTP	665	HTTP/1.1 200 OK (JPEG 7F1F image)

Sequence Number: 4533 (relative sequence number)
Sequence Number (raw): 1493726126
[Next Sequence Number: 4915 (relative sequence number)]
Acknowledgment Number: 20671 (relative ack number)
Acknowledgment number (raw): 1617146744
1000 = Header Length: 32 bytes (8)
Flags: 0x018 (PSH, ACK)
Window: 65535
[Calculated window size: 524288]
[Window size scaling factor: 8]
Checksum: 0x63e3 [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
[Timestamps]
[SEQ/ACK analysis]
TCP payload (382 bytes)
Hypertext Transfer Protocol
GET /images/G/01/stores/sport-goods/backcountry-womenapp_V255873153_.jpg HTTP/1.1\r\n
User-Agent: Mozilla/5.0 (Macintosh; U; Intel Mac OS X 10_5_4; en-us) AppleWebKit/525.18 (KHTML, like Gecko) Version/3.1.2 Safari/525.20.1\r\n
Referer: http://www.amazon.com/\r\n
Accept: */*\r\n
Accept-Language: en-us\r\n
Accept-Encoding: gzip, deflate\r\n
Connection: keep-alive\r\n
Host: g-ecx.images-amazon.com\r\n
\r\n
[response in frame 19568]
[Full request URI: http://g-ecx.images-amazon.com/images/G/01/stores/sport-goods/backcountry-womenapp_V255873153_.jpg]
HTTP Referer (http.referer), 33 bytes
Packets: 94410 - Displayed: 9394 (10.0%)
Profile: Default

Evidence 03

The screenshot shows the Wireshark interface with the filter 'ip.src==140.247.62.34 or ip.dst==140.247.62.34' applied. The packet list displays a series of TCP connections between 140.247.62.34 and 192.168.15.4. The packet details pane shows the structure of a TCP segment, including the header, options, and the payload (a 78-byte frame). The packet bytes pane shows the raw data in hexadecimal and ASCII.

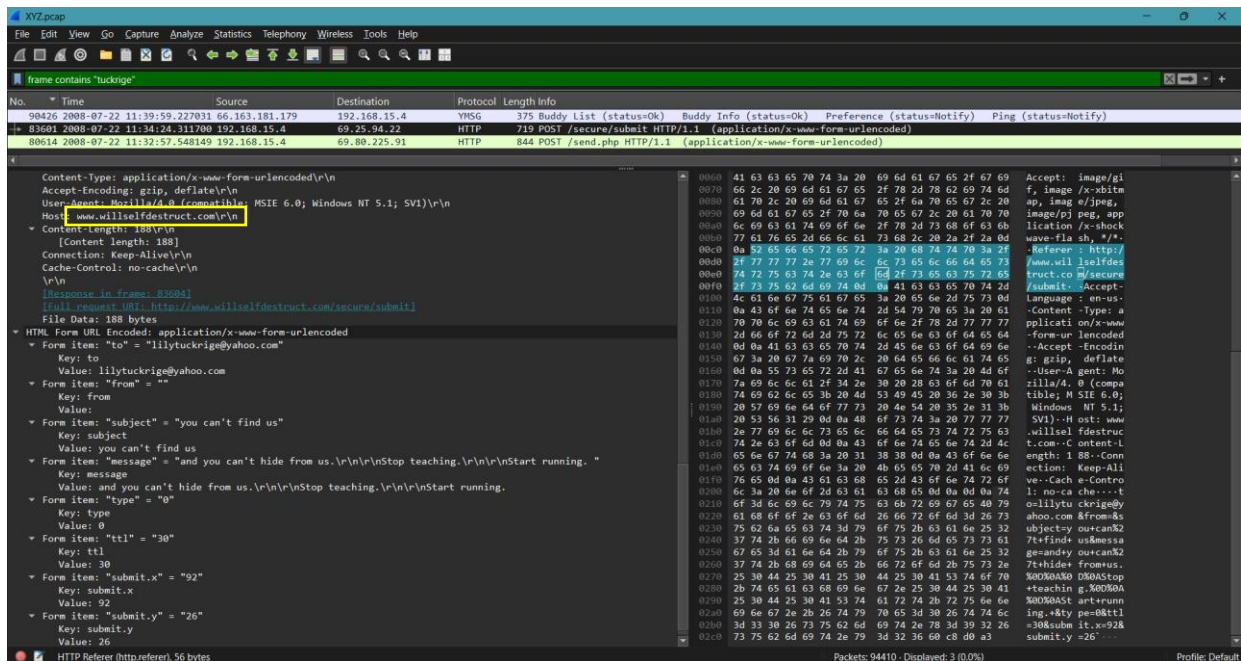
Description: Filtering the XYZ.pcap using the IP address used to send the harassing email by the filter `ip.src == 140.247.62.34 or ip.dst == 140.247.62.34`. When look at the IP address source and destination, can see the IP address 192.168.15.4 which is the only IP bridging with IP 140.247.62.34. It is the public IP address of the router. When look at the MAC addresses of each IP addresses in the Data Link layer, can find the hardware of each device.

Evidence 04

The screenshot shows the Wireshark interface with the filter 'frame contains "tuckrige"' applied. The packet list displays three packets, with the last two being HTTP requests. The packet details pane shows the structure of an HTTP POST request, including the header, body, and the payload (a 774-byte frame). The packet bytes pane shows the raw data in hexadecimal and ASCII.

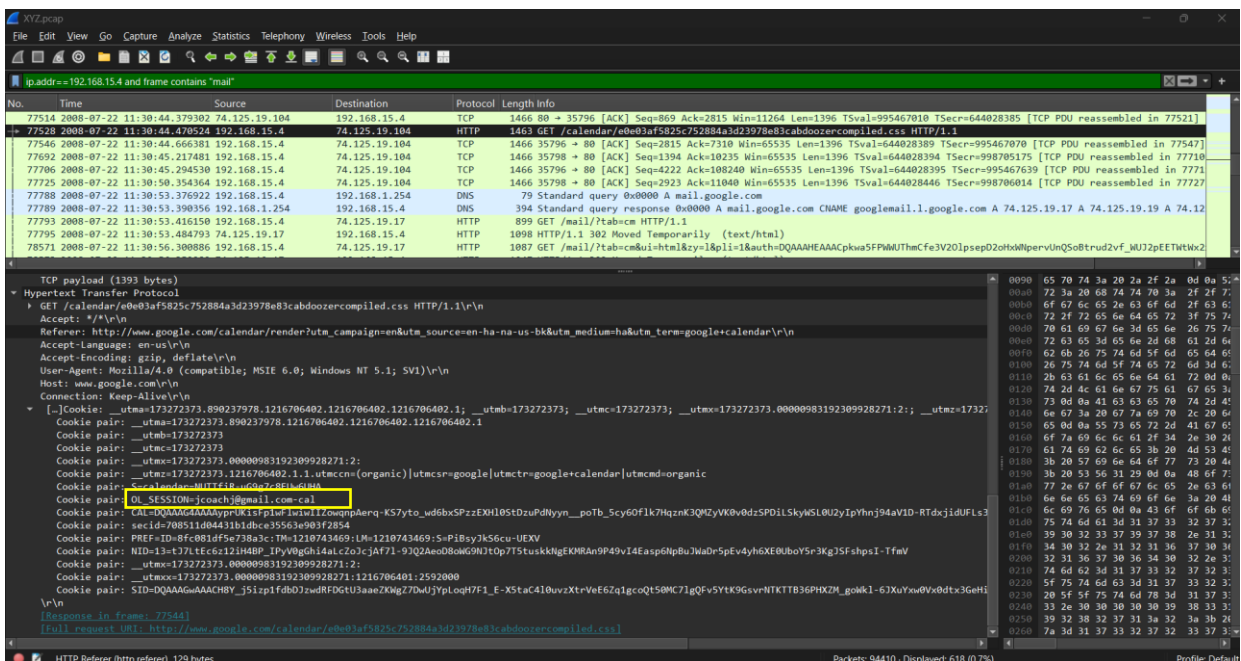
Description: When filtering the pcap file using the filter `frame contains "tuckrige"`, can see 3 packets where last two packets use HTTP. The packet number 80614 shows a harassing message sent using `sendanonymousemail.net`. The source and destination IPs are 192.168.15.4 and 69.80.225.91 respectively.

Evidence 05



Description: We can see that the packet number 83601 shows that the harassing message was sent using Willselfdestruct.com. Here, the source IP is 192.168.15.4 and destination IP is 69.25.94.22

Evidence 06



Description: We further looked into the packets by filtering using `ip.addr == 192.168.15.4` and frame contains "mail" to find the email address of the attacker. The packet number 77528 contains an email address jcoachj@gmail.com.

Evidence 07

The screenshot shows the Wireshark Statistics - Conversations tab. The 'Conversations' list on the left is filtered by 'Ethernet II'. The main table displays the following data:

Address A	Address B	Packets	Bytes	Stream ID	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel. Start	Duration	Bits/s A → B	Bits/s B → A
Apple_5a:77:9b	Apple_69:38:cc	2	128 bytes	11	2	128 bytes	0	0 bytes	4380.593992	301.1980	3 bits/s	0 bits/s
Apple_5a:77:9b	Broadcast	124	8 kB	8	124	8 kB	0	0 bytes	4377.940322	975.4316	65 bits/s	0 bits/s
Apple_5a:77:9b	CommScope_99:98:68	46	8 kB	13	35	2 kB	11	6 kB	4680.609544	803.4276	22 bits/s	53 bits/s
Apple_5a:77:9b	IPv4mcast_02	5	320 bytes	9	5	320 bytes	0	0 bytes	4379.563606	303.3748	8 bits/s	0 bits/s
Apple_5a:77:9b	IPv4mcast_fb	10	640 bytes	10	10	640 bytes	0	0 bytes	4379.563612	977.2436	5 bits/s	0 bits/s
Apple_69:38:cc	Broadcast	38	2 kB	6	38	2 kB	0	0 bytes	4314.010482	5203.3383	2 bits/s	0 bits/s
Apple_69:38:cc	CommScope_99:98:68	12	504 bytes	2	12	504 bytes	0	0 bytes	126.365108	6946.9978	0 bits/s	0 bits/s
Apple_69:38:cc	HonHaiPrecis_2e:4f:60	7	294 bytes	27	7	294 bytes	0	0 bytes	5941.224417	5408.8345	0 bits/s	0 bits/s
Apple_69:38:cc	IPv4mcast_02	2	92 bytes	16	2	92 bytes	0	0 bytes	6539.241610	2984.9993	0 bits/s	0 bits/s
Apple_69:38:cc	IPv4mcast_fb	14	644 bytes	7	14	644 bytes	0	0 bytes	4317.620969	5215.6242	0 bits/s	0 bits/s
Apple_79:00:31	Broadcast	54	4 kB	24	54	4 kB	0	0 bytes	9768.817010	219.1614	128 bits/s	0 bits/s
Apple_79:00:31	IPv4mcast_02	2	128 bytes	26	2	128 bytes	0	0 bytes	9906.797886	80.2287	12 bits/s	0 bits/s
Apple_79:00:31	IPv4mcast_fb	7	448 bytes	25	7	448 bytes	0	0 bytes	9771.442764	221.0956	16 bits/s	0 bits/s
Apple_b4:a3:f8	Apple_e2:c0:ce	8	512 bytes	29	8	512 bytes	0	0 bytes	15640.853401	1.6231	2523 bits/s	0 bits/s
Apple_b4:a3:f8	Broadcast	15	1 kB	28	15	1 kB	0	0 bytes	15640.663866	116.8263	80 bits/s	0 bits/s
Apple_b4:a3:f8	IPv4mcast_fb	2	128 bytes	30	2	128 bytes	0	0 bytes	15646.104324	4.9761	205 bits/s	0 bits/s
Apple_e2:c0:ce	Broadcast	21	1 kB	18	21	1 kB	0	0 bytes	9524.561107	6049.1290	1 bits/s	0 bits/s
Apple_f1:8a:6e	Broadcast	5	320 bytes	15	5	320 bytes	0	0 bytes	5763.806879	8.3449	306 bits/s	0 bits/s
Apple_f1:8a:6e	CommScope_99:98:68	1	77 bytes	5	1	77 bytes	0	0 bytes	1311.733266	0.0000	0 bits/s	0 bits/s
CommScope_99:98:68	Broadcast	3,482	223 kB	1	3,482	223 kB	0	0 bytes	126.345039	6949.5184	256 bits/s	0 bits/s
CommScope_99:98:68	IPv4mcast_7fff:fa	66	22 kB	3	66	22 kB	0	0 bytes	214.663047	6844.1773	26 bits/s	0 bits/s
HonHaiPrecis_2e:4f:60	Apple_e2:c0:ce	73,225	45 MB	19	38,664	59 MB	34,561	6 MB	9524.562146	6235.3686	50 kbps	7358 bits/s
HonHaiPrecis_2e:4f:60	Apple_f1:8a:6e	5	459 bytes	12	4	382 bytes	1	77 bytes	4462.790144	866.4946	3 bits/s	0 bits/s
HonHaiPrecis_2e:4f:60	Broadcast	43	3 kB	20	43	3 kB	0	0 bytes	9526.156244	6120.1662	3 bits/s	0 bits/s
HonHaiPrecis_2e:4f:60	IPv4mcast_7fff:fa	2,150	795 kB	14	2,150	795 kB	0	0 bytes	5288.759237	10469.0989	607 bits/s	0 bits/s
HonHaiPrecis_2e:4f:60	Broadcast	98	6 kB	4	98	6 kB	0	0 bytes	342.775583	15323.2692	3 bits/s	0 bits/s
HonHaiPrecis_2e:4f:60	CommScope_99:98:68	14,892	9 MB	0	6,818	1 MB	8,074	8 MB	0.000000	7154.7393	1215 bits/s	8676 bits/s
TRENNet_44:a0:f1	Broadcast	61	4 kB	17	61	4 kB	0	0 bytes	9516.170232	6184.1372	5 bits/s	0 bits/s
TRENNet_44:a0:f1	IPv4mcast_09	1	70 bytes	22	1	70 bytes	0	0 bytes	9529.936674	0.0000	0 bits/s	0 bits/s
TRENNet_44:a0:f1	IPv4mcast_16	3	192 bytes	21	3	192 bytes	0	0 bytes	9529.934162	0.9794	1568 bits/s	0 bits/s
TRENNet_44:a0:f1	IPv4mcast_7fff:fa	9	1 kB	23	9	1 kB	0	0 bytes	9529.945028	1568.5768	7 bits/s	0 bits/s

Description: Conversations tab under Statistics listing all MAC addresses detected on the Network. This was obtained using Wireshark ver4.4.3. There are 17 distinct Ethernet components. The most active ones out of the above devices are Apple_e2:c0:ce and HonHaiPrecis_2e:4f:60. They are exchanging the highest amount of packets compared to other devices.

Evidence 8

The screenshot shows the Wireshark Statistics - Conversations tab. The 'Conversations' list on the left is filtered by 'IPv4'. The main table displays the following data:

Address A	Address B	Packets	Bytes	Stream ID	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel. Start	Duration	Bits/s A → B	Bits/s B → A
10.0.1.5	10.0.1.255	2	204 bytes	201	2	204 bytes	0	0 bytes	9929.465347	1.1264	1448 bits/s	0 bits/s
10.0.1.5	224.0.0.2	1	64 bytes	200	1	64 bytes	0	0 bytes	9906.797886	0.0000	0 bits/s	0 bits/s
10.0.1.5	224.0.0.251	5	320 bytes	183	5	320 bytes	0	0 bytes	9771.442764	141.4877	18 bits/s	0 bits/s
24.64.79.171	192.168.1.64	3	2 kB	77	3	2 kB	0	0 bytes	4686.427360	0.0022	0 bits/s	0 bits/s
24.64.124.214	192.168.1.64	3	2 kB	59	3	2 kB	0	0 bytes	681.438457	0.0018	0 bits/s	0 bits/s
24.64.120.46	192.168.1.64	3	2 kB	66	3	2 kB	0	0 bytes	948.426395	0.0016	0 bits/s	0 bits/s
24.190.178.195	192.168.1.64	1	144 bytes	100	1	144 bytes	0	0 bytes	5782.520884	0.0000	0 bits/s	0 bits/s
38.107.163.7	192.168.1.64	5	390 bytes	98	5	390 bytes	0	0 bytes	5778.857577	44.9983	69 bits/s	0 bits/s
58.8.102.59	192.168.1.64	3	234 bytes	83	3	234 bytes	0	0 bytes	5759.173770	8.9974	208 bits/s	0 bits/s
60.54.96.180	192.168.1.64	1	144 bytes	91	1	144 bytes	0	0 bytes	5767.918200	0.0000	0 bits/s	0 bits/s
60.222.224.130	192.168.1.64	1	503 bytes	58	1	503 bytes	0	0 bytes	582.167223	0.0000	0 bits/s	0 bits/s
60.222.224.141	192.168.1.64	1	503 bytes	91	1	503 bytes	0	0 bytes	7023.233273	0.0000	0 bits/s	0 bits/s
60.222.245.13	192.168.1.64	1	144 bytes	97	1	144 bytes	0	0 bytes	5776.221934	0.0000	0 bits/s	0 bits/s
60.222.253.104	192.168.1.64	2	1 kB	62	2	1 kB	0	0 bytes	765.710608	0.0006	0 bits/s	0 bits/s
61.16.70.132	192.168.1.64	1	147 bytes	125	1	147 bytes	0	0 bytes	5827.184259	0.0000	0 bits/s	0 bits/s
63.247.440.161	192.168.1.64	80	6 kB	231	13	2 kB	67	6 kB	10624.122767	4328.3817	3 bits/s	11 bits/s
65.125.141.18	192.168.1.64	2	140 bytes	88	2	140 bytes	0	0 bytes	5766.838127	2.9891	374 bits/s	0 bits/s
66.114.51.42	192.168.1.64	11	770 bytes	228	1	70 bytes	10	700 bytes	10615.811314	214.4692	2 bits/s	26 bits/s
66.151.152.143	192.168.1.64	1	64 bytes	475	1	64 bytes	0	0 bytes	15560.227108	0.0000	0 bits/s	0 bits/s
66.228.249.145	192.168.1.64	2	132 bytes	123	2	132 bytes	0	0 bytes	5824.779545	2.9016	363 bits/s	0 bits/s
67.85.10.95	192.168.1.64	1	144 bytes	86	1	144 bytes	0	0 bytes	5761.818609	0.0000	0 bits/s	0 bits/s
67.142.33.16	192.168.1.64	1	147 bytes	90	1	147 bytes	0	0 bytes	5767.292663	0.0000	0 bits/s	0 bits/s
69.39.67.98	192.168.1.64	80	7 kB	227	13	1 kB	67	5 kB	10613.629181	4338.9841	2 bits/s	9 bits/s
69.59.23.129	192.168.1.64	1,072	232 kB	138	494	107 kB	578	125 kB	6636.450310	11.3446	75 kbps	87 kbps
70.50.251.14	192.168.1.64	1	66 bytes	142	1	66 bytes	0	0 bytes	6746.885587	0.0000	0 bits/s	0 bits/s
70.72.57.167	192.168.1.64	1	147 bytes	117	1	147 bytes	0	0 bytes	5818.437232	0.0000	0 bits/s	0 bits/s
71.209.246.42	192.168.1.64	1	149 bytes	99	1	149 bytes	0	0 bytes	5779.949777	0.0000	0 bits/s	0 bits/s
71.237.122.85	192.168.1.64	1	149 bytes	120	1	149 bytes	0	0 bytes	5821.435660	0.0000	0 bits/s	0 bits/s
74.66.229.33	192.168.1.64	3	198 bytes	118	3	198 bytes	0	0 bytes	5819.537063	8.6035	184 bits/s	0 bits/s
75.152.226.9	192.168.1.64	1	147 bytes	110	1	147 bytes	0	0 bytes	5800.991063	0.0000	0 bits/s	0 bits/s
79.131.4.159	192.168.1.64	1	149 bytes	114	1	149 bytes	0	0 bytes	5810.524900	0.0000	0 bits/s	0 bits/s
81.57.107.160	192.168.1.64	1	149 bytes	92	1	149 bytes	0	0 bytes	5769.913052	0.0000	0 bits/s	0 bits/s
81.157.31.100	192.168.1.64	1	144 bytes	95	1	144 bytes	0	0 bytes	5773.703276	0.0000	0 bits/s	0 bits/s
84.76.221.226	192.168.1.64	1	144 bytes	115	1	144 bytes	0	0 bytes	5813.109775	0.0000	0 bits/s	0 bits/s
87.990.220	192.168.1.64	2	288 bytes	87	2	288 bytes	0	0 bytes	5765.580926	5.7985	397 bits/s	0 bits/s
87.121.25.26	192.168.1.64	1	147 bytes	85	1	147 bytes	0	0 bytes	5759.828091	0.0000	0 bits/s	0 bits/s
89.1128.81	192.168.1.64	2	132 bytes	122	2	132 bytes	0	0 bytes	5823.978235	6.0880	173 bits/s	0 bits/s
89.216.64.93	192.168.1.64	1	66 bytes	84	1	66 bytes	0	0 bytes	5759.727529	0.0000	0 bits/s	0 bits/s
90.26.122.118	192.168.1.64	1	149 bytes	103	1	149 bytes	0	0 bytes	5786.356405	0.0000	0 bits/s	0 bits/s
91.65.135.197	192.168.1.64	1	144 bytes	116	1	144 bytes	0	0 bytes	5814.481826	0.0000	0 bits/s	0 bits/s
93.103.36.196	192.168.1.64	1	149 bytes	112	1	149 bytes	0	0 bytes	5800.527232	0.0000	0 bits/s	0 bits/s
98.220.46.34	192.168.1.64	1	149 bytes	104	1	149 bytes	0	0 bytes	5787.622977	0.0000	0 bits/s	0 bits/s
116.252.234.84	192.168.1.64	1	147 bytes	109	1	147 bytes	0	0 bytes	5800.174858	0.0000	0 bits/s	0 bits/s

Description: Conversations tab under Statistics listing all IPv4 addresses detected on the Network. This was obtained using Wireshark ver4.4.3. Multiple IPs were detected of which 6 belong to the network that was monitored under the 192.168.15.xx subnet.

Evidence 09

Bluetooth	Ethernet - 31	IPv4 - 482	IPv6	TCP - 2027	UDP - 828	Stream ID	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel. Start	Duration	Bits/s A → B	Bits/s B → A
	Address A	Port A Address B	Port B	Packets	Bytes									
	192.168.15.4	36808 12.130.60.2	80	23	12 kb	1999	12	2 kb	11	10 kb	15563.437824	0.7399	26 kbps	105 kbps
	192.168.15.4	36542 12.130.60.2	80	23	12 kb	1876	12	2 kb	11	10 kb	15533.691875	0.7447	25 kbps	105 kbps
	192.168.15.4	36410 12.130.60.2	80	23	12 kb	1816	12	2 kb	11	10 kb	15508.284022	0.6903	27 kbps	113 kbps
	192.168.15.4	35256 12.130.60.2	80	17	8 kb	1380	9	2 kb	8	7 kb	14548.139528	0.4714	26 kbps	114 kbps
	192.168.15.4	35228 12.130.60.2	80	17	8 kb	1368	9	2 kb	8	7 kb	14507.583929	0.4032	31 kbps	133 kbps
	192.168.15.4	35188 12.130.60.2	80	17	9 kb	1350	9	2 kb	8	7 kb	14477.894588	0.4496	28 kbps	125 kbps
	192.168.15.4	35168 12.130.60.2	80	17	9 kb	1341	9	2 kb	8	7 kb	14450.959718	0.5038	25 kbps	113 kbps
	192.168.15.4	32946 12.130.60.2	443	33	15 kb	447	17	2 kb	16	12 kb	9850.611160	1.1366	17 kbps	86 kbps
	192.168.15.4	36076 12.129.210.46	80	10	2 kb	1723	6	1 kb	4	1 kb	15227.577608	1.8039	4833 bits/s	6239 bits/s
	192.168.15.4	35944 12.129.210.46	80	30	10 kb	1664	24	5 kb	6	5 kb	15130.094544	588.2796	62 bits/s	71 bits/s
	192.168.15.4	35652 12.129.210.41	80	20	3 kb	1546	18	2 kb	2	1 kb	14845.470330	530.3761	27 bits/s	18 bits/s
	192.168.15.4	35644 12.129.210.41	80	5	3 kb	1541	3	2 kb	2	608 bytes	14832.869997	12.6015	1218 bits/s	385 bits/s
	192.168.15.4	35084 12.129.147.65	80	10	1 kb	1307	6	940 bytes	4	472 bytes	14410.761574	1.2737	5903 bits/s	2964 bits/s
	192.168.15.4	35082 12.129.147.65	80	10	1 kb	1306	6	937 bytes	4	463 bytes	14410.761261	1.2744	5882 bits/s	2906 bits/s
	192.168.15.4	34482 12.129.147.65	80	10	1 kb	1061	6	940 bytes	4	472 bytes	12608.592051	1.3404	5610 bits/s	2817 bits/s
	192.168.15.4	34480 12.129.147.65	80	10	1 kb	1060	6	937 bytes	4	463 bytes	12608.591634	1.3405	5591 bits/s	2783 bits/s
	192.168.15.4	33320 12.129.147.65	80	10	1 kb	604	6	940 bytes	4	472 bytes	10807.418848	1.3616	5522 bits/s	2773 bits/s
	192.168.15.4	33318 12.129.147.65	80	10	1 kb	603	6	937 bytes	4	463 bytes	10807.418365	1.3615	5505 bits/s	2720 bits/s
	192.168.15.4	33298 8.12.221.123	80	13	4 kb	594	7	2 kb	6	1 kb	10796.550420	18.2527	1052 bits/s	572 bits/s
	192.168.15.4	32830 8.12.217.125	80	60	27 kb	390	36	5 kb	24	22 kb	9528.825132	542.3640	71 bits/s	320 bits/s
	192.168.15.4	32828 8.12.217.125	80	71	26 kb	389	41	6 kb	30	20 kb	9528.824713	542.3647	87 bits/s	296 bits/s
	192.168.15.4	32824 8.12.217.125	80	87	35 kb	387	52	9 kb	35	26 kb	9528.286800	542.9029	126 bits/s	389 bits/s
	192.168.15.4	32822 8.12.217.125	80	235	160 kb	386	108	15 kb	127	145 kb	9528.286259	542.9038	218 bits/s	2142 bits/s
	192.168.15.4	35712 4.78.212.29	80	8	1 kb	1574	5	690 bytes	3	684 bytes	14893.351437	0.2669	20 kbps	20 kbps
	192.168.15.4	35458 4.78.212.29	80	8	1 kb	1459	5	774 bytes	3	684 bytes	14777.257022	0.2603	23 kbps	21 kbps
	192.168.15.4	33950 4.71.104.187	80	19	8 kb	1667	12	4 kb	7	4 kb	15130.016745	91.6796	328 bits/s	353 bits/s
	192.168.1.64	49019 216.246.105.11	8020	10	1 kb	379	7	902 bytes	3	226 bytes	7136.579567	18.1997	397 bits/s	99 bits/s
	192.168.1.64	35169 216.246.105.11	8020	25	3 kb	378	16	2 kb	9	632 bytes	7116.430606	31.8044	545 bits/s	158 bits/s
	192.168.1.64	43723 216.239.122.151	80	15	6 kb	316	8	806 bytes	7	5 kb	6585.440554	198.8287	32 bits/s	193 bits/s
	192.168.1.64	43507 216.239.122.151	80	19	6 kb	372	10	833 bytes	9	5 kb	6938.349202	34.1327	195 bits/s	1158 bits/s
	192.168.1.64	32972 216.239.122.151	80	21	10 kb	270	9	998 bytes	12	9 kb	395.550812	38.6360	205 bits/s	1933 bits/s
	192.168.1.64	43112 212.58.226.79	80	30	21 kb	348	12	1 kb	18	19 kb	6750.495489	14.9366	765 bits/s	10 kbps
	192.168.1.64	34763 212.58.226.75	80	29	20 kb	15	12	1 kb	17	19 kb	17.319278	13.4120	852 bits/s	11 kbps
	192.168.1.64	38912 209.225.0.101	80	12	4 kb	265	6	2 kb	6	2 kb	315.895614	0.3044	43 kbps	51 kbps
	192.168.1.64	48490 209.85.201.189	80	11	2 kb	358	6	2 kb	5	628 bytes	6824.484799	16.3404	862 bits/s	307 bits/s
	192.168.1.64	46202 209.85.171.97	443	23	5 kb	6	10	2 kb	13	2 kb	2.200937	131.1012	134 bits/s	184 bits/s
	192.168.1.64	38913 209.85.171.97	443	27	7 kb	4	12	3 kb	15	4 kb	1.309339	140.7561	167 bits/s	226 bits/s
	192.168.1.64	46202 209.3.183.2	80	18	6 kb	10	11	2 kb	7	4 kb	12.164453	7.5104	1889 bits/s	4329 bits/s
	192.168.1.64	44565 209.3.183.2	80	41	32 kb	11	16	2 kb	25	30 kb	12.165119	7.5100	2265 bits/s	31 kbps
	192.168.1.64	44018 209.3.183.2	80	37	28 kb	8	14	2 kb	23	26 kb	12.162845	7.5113	2123 bits/s	27 kbps
	192.168.1.64	34871 209.3.183.2	80	35	26 kb	9	14	2 kb	21	24 kb	12.163746	7.5108	2120 bits/s	25 kbps
	192.168.1.64	37314 208.111.153.175	80	33	21 kb	309	15	2 kb	18	19 kb	6548.550670	0.0778	187 kbps	2000 kbps

Description: Conversations tab under Statistics shows all TCP components captured on the network which was obtained using Wireshark ver4.4.3. This image highlights that there are several TCP transactions for the IP addresses 192.168.15.4 and 192.168.1.64. We can see that there are several TCP transactions for the suspect IP 192.168.15.4 on port numbers 80 and 443.

Evidence 010

Wireshark - Conversations - XYZ.pcap

Conversation Settings

Bluetooth

Ethernet - 31

IPv4 - 482

IPv6

TCP - 2027

UDP - 828

Name resolution

Absolute start time

Limit to display filter

Copy

Follow Stream...

Graph...

Protocol

☒ Bluetooth

☐ SPX

☐ DCCP

☒ Ethernet

☐ FC

☐ FDDI

☐ IEEE 802.11

☐ IEEE 802.15.4

☐ IPv4

☒ IPv6

☐ JXTA

☐ LTP

☐ MPTCP

☐ NCP

☐ openSAFETY

☐ RSP

☐ SCTP

☐ SLL

☒ TCP

Filter list for specific type

Address A	Port A	Address B	Port B	Packets	Bytes	Stream ID	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel. Start	Duration	Bits/s A → B	Bits/s B → A
192.168.15.4	3561	192.168.1.254	53	2	152 bytes	497	1	76 bytes	1	76 bytes	10582.106303	0.0146	41 kbps	41 kbps
192.168.15.4	3560	192.168.1.254	53	2	166 bytes	496	1	83 bytes	1	83 bytes	10582.030269	0.1030	6444 bits/s	6444 bits/s
192.168.15.4	3559	192.168.1.254	53	2	150 bytes	495	1	75 bytes	1	75 bytes	10582.007616	0.0132	45 kbps	45 kbps
192.168.15.4	3558	192.168.1.254	53	2	151 bytes	494	1	75 bytes	1	376 bytes	10582.006001	0.0137	43 kbps	219 kbps
192.168.15.4	3557	192.168.1.254	53	2	152 bytes	493	1	76 bytes	1	76 bytes	10581.940961	0.0128	44 kbps	44 kbps
192.168.15.4	3556	192.168.1.254	53	2	158 bytes	492	1	79 bytes	1	79 bytes	10577.749473	0.0207	30 kbps	30 kbps
192.168.15.4	3555	192.168.1.254	53	2	158 bytes	491	1	79 bytes	1	79 bytes	10572.687049	0.0136	46 kbps	46 kbps
192.168.15.4	3554	192.168.1.254	53	2	167 bytes	490	1	79 bytes	1	388 bytes	10572.685582	0.0140	45 kbps	222 kbps
192.168.15.4	3553	192.168.1.254	53	2	164 bytes	489	1	78 bytes	1	386 bytes	10571.811163	0.0140	44 kbps	220 kbps
192.168.15.4	3552	192.168.1.254	53	2	170 bytes	488	1	89 bytes	1	89 bytes	10562.399847	0.0061	18 kbps	18 kbps
192.168.15.4	3551	192.168.1.254	53	2	156 bytes	487	1	78 bytes	1	78 bytes	10561.990383	0.0967	6455 bits/s	6455 bits/s
192.168.15.4	3540	192.168.1.254	53	2	158 bytes	486	1	79 bytes	1	79 bytes	10497.918018	0.0137	46 kbps	46 kbps
192.168.15.4	3539	192.168.1.254	53	2	162 bytes	485	1	81 bytes	1	81 bytes	10497.256248	0.0176	36 kbps	36 kbps
192.168.15.4	3538	192.168.1.254	53	2	500 bytes	484	1	81 bytes	1	419 bytes	10497.254130	0.0184	35 kbps	181 kbps
192.168.15.4	3537	192.168.1.254	53	2	184 bytes	483	1	82 bytes	1	82 bytes	10497.240600	0.0178	7526 bits/s	7526 bits/s
192.168.15.4	3536	192.168.1.254	53	2	162 bytes	482	1	81 bytes	1	81 bytes	10369.512327	0.0322	20 kbps	20 kbps
192.168.15.4	3535	192.168.1.254	53	2	158 bytes	481	1	79 bytes	1	79 bytes	10368.271849	0.0133	47 kbps	47 kbps
192.168.15.4	3534	192.168.1.254	53	2	172 bytes	480	1	86 bytes	1	86 bytes	10362.049984	0.1017	6763 bits/s	6763 bits/s
192.168.15.4	3533	192.168.1.254	53	2	186 bytes	479	1	93 bytes	1	93 bytes	10362.023802	0.1024	7263 bits/s	7263 bits/s
192.168.15.4	3532	192.168.1.254	53	2	172 bytes	478	1	86 bytes	1	86 bytes	10362.022348	0.1033	6660 bits/s	6660 bits/s
192.168.15.4	3531	192.168.1.254	53	2	156 bytes	477	1	78 bytes	1	78 bytes	10362.020867	0.0175	35 kbps	35 kbps
192.168.15.4	3530	192.168.1.254	53	2	172 bytes	476	1	78 bytes	1	494 bytes	10362.019511	0.0162	38 kbps	243 kbps
192.168.15.4	3529	192.168.1.254	53	2	172 bytes	475	1	86 bytes	1	86 bytes	10362.018215	0.1020	6744 bits/s	6744 bits/s
192.168.15.4	3528	192.168.1.254	53	2	170 bytes	474	1	85 bytes	1	85 bytes	10361.977797	0.0197	34 kbps	34 kbps
192.168.15.4	3527	192.168.1.254	53	2	562 bytes	473	1	85 bytes	1	472 bytes	10361.976273	0.0207	32 kbps	184 kbps
192.168.15.4	3526	192.168.1.254	53	2	156 bytes	472	1	78 bytes	1	78 bytes	10361.974676	0.0196	31 kbps	31 kbps
192.168.15.4	3525	192.168.1.254	53	2	532 bytes	471	1	78 bytes	1	454 bytes	10361.973182	0.0201	31 kbps	180 kbps
192.168.15.4	3524	192.168.1.254	53	2	156 bytes	470	1	78 bytes	1	78 bytes	10361.970413	0.0141	44 kbps	44 kbps
192.168.15.4	3523	192.168.1.254	53	2	532 bytes	469	1	78 bytes	1	454 bytes	10361.969038	0.0145	43 kbps	250 kbps
192.168.15.4	3522	192.168.1.254	53	2	532 bytes	468	1	74 bytes	1	478 bytes	10361.962694	0.0203	29 kbps	187 kbps
192.168.15.4	3521	192.168.1.254	53	2	508 bytes	467	1	87 bytes	1	421 bytes	10360.007067	0.0154	45 kbps	219 kbps
192.168.15.4	3520	192.168.1.254	53	2	156 bytes	466	1	78 bytes	1	78 bytes	10359.481259	0.0202	30 kbps	30 kbps
192.168.15.4	3519	192.168.1.254	53	2	154 bytes	465	1	77 bytes	1	77 bytes	10359.479714	0.0201	30 kbps	30 kbps
192.168.15.4	3517	192.168.1.254	53	2	174 bytes	464	1	87 bytes	1	87 bytes	10195.887411	0.0137	50 kbps	50 kbps
192.168.15.4	3516	192.168.1.254	53	2	174 bytes	463	1	87 bytes	1	87 bytes	10195.876537	0.0133	52 kbps	52 kbps
192.168.15.4	3515	192.168.1.254	53	2	172 bytes	462	1	86 bytes	1	86 bytes	10195.867162	0.1062	6477 bits/s	6477 bits/s
192.168.15.4	3514	192.168.1.254	53	2	174 bytes	461	1	87 bytes	1	87 bytes	10195.856821	0.1032	6741 bits/s	6741 bits/s
192.168.15.4	3504	192.168.1.254	53	2	162 bytes	460	1	87 bytes	1	87 bytes	10149.858993	0.2269	1982 bits/s	1982 bits/s
192.168.15.4	3503	192.168.1.254	53	2	231 bytes	459	1	79 bytes	1	152 bytes	10145.498427	0.0143	44 kbps	84 kbps
192.168.15.4	3501	192.168.1.254	53	2	174 bytes	458	1	87 bytes	1	87 bytes	9973.719478	0.1012	6874 bits/s	6874 bits/s
192.168.15.4	3500	192.168.1.254	53	2	174 bytes	457	1	87 bytes	1	87 bytes	9973.709309	0.1000	6958 bits/s	6958 bits/s
192.168.15.4	3499	192.168.1.254	53	2	172 bytes	456	1	86 bytes	1	86 bytes	9973.699281	0.1012	6796 bits/s	6796 bits/s
192.168.15.4	3498	192.168.1.254	53	2	174 bytes	455	1	87 bytes	1	87 bytes	9973.689148	0.1008	6907 bits/s	6907 bits/s

Close Help

Evidence 011

Wireshark - HTTP / Load Distribution - XYZ.pcap

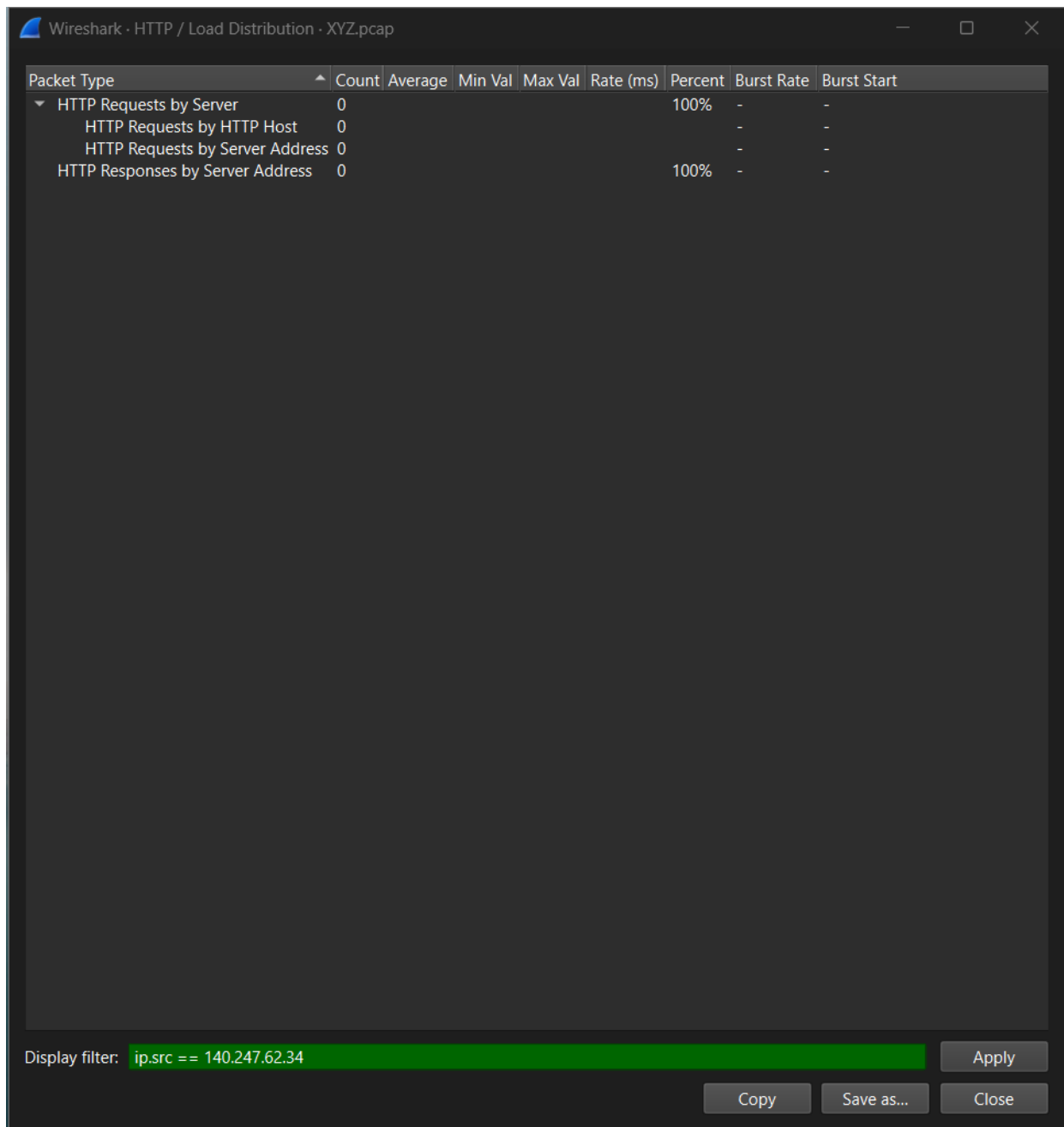
Packet Type	Count	Average	Min Val	Max Val	Rate (ms)	Percent	Burst Rate	Burst Start
HTTP Requests by Server	4841				0.0003	100%	0.1800	9850.228
HTTP Requests by HTTP Host	4841				0.0003	100.00%	0.1800	9850.228
0.channel14.facebook.com	6				0.0000	0.12%	0.0100	10773.733
69.63.176.174	6				0.0000	100.00%	0.0100	10773.733
0.channel14.facebook.com:80	1				0.0000	0.02%	0.0100	10771.760
69.63.176.174	1				0.0000	100.00%	0.0100	10771.760
a.dlqm.net	3				0.0000	0.06%	0.0100	15140.328
12.130.81.249	3				0.0000	100.00%	0.0100	15140.328
a1.phobos.apple.com	122				0.0000	2.52%	0.1000	15558.765
69.22.167.199	5				0.0000	4.10%	0.0400	15526.681
69.22.167.201	67				0.0000	54.92%	0.1000	15558.765
69.22.167.214	20				0.0000	16.39%	0.0100	10966.653
69.22.167.215	30				0.0000	24.59%	0.0400	106.325
a1.reunion.com	3				0.0000	0.06%	0.0300	14817.975
69.22.167.248	3				0.0000	100.00%	0.0300	14817.975
about.dtmplib.com	1				0.0000	0.02%	0.0100	15123.164
208.185.127.40	1				0.0000	100.00%	0.0100	15123.164
ad.doubleclick.net	5				0.0000	0.10%	0.0200	9529.687
216.73.86.152	1				0.0000	20.00%	0.0100	11334.386
216.73.86.52	2				0.0000	40.00%	0.0200	9529.687
216.73.87.52	1				0.0000	20.00%	0.0100	9855.202
65.205.8.52	1				0.0000	20.00%	0.0100	11184.153
ad.yieldmanager.com	41				0.0000	0.85%	0.0200	11183.017
74.6.104.11	4				0.0000	9.76%	0.0100	14450.945
76.13.208.11	13				0.0000	31.71%	0.0200	14817.244
76.13.212.11	24				0.0000	58.54%	0.0200	11183.017
address.yahoo.com	2				0.0000	0.04%	0.0100	11458.606
209.191.93.51	2				0.0000	100.00%	0.0100	11458.606
adopt.specificklick.net	4				0.0000	0.08%	0.0100	9530.143
64.79.161.90	4				0.0000	100.00%	0.0100	9530.143
ads.ak.facebook.com	14				0.0000	0.29%	0.0100	10771.253
69.22.167.247	1				0.0000	7.14%	0.0100	10771.253
69.26.180.23	13				0.0000	92.86%	0.0100	10797.575
ads.cnn.com	9				0.0000	0.19%	0.0300	9850.151
64.236.22.103	9				0.0000	100.00%	0.0300	9850.151
ads.cnn.com:80	4				0.0000	0.08%	0.0100	9850.473
64.236.22.103	4				0.0000	100.00%	0.0100	9850.473
ads.yimg.com	2				0.0000	0.04%	0.0200	15403.984
69.22.167.224	2				0.0000	100.00%	0.0200	15403.984
ads1.msn.com	1				0.0000	0.02%	0.0100	10811.209
199.93.58.125	1				0.0000	100.00%	0.0100	10811.209
afe.specificklick.net	4				0.0000	0.08%	0.0100	9529.971
200.95.66.221	4				0.0000	100.00%	0.0100	9529.971

Display filter: http

Copy Save as... Close

Description: HTTP load distribution pattern with http filter on the Statistics tab listing all TCP components detected on the Network. This was obtained on the HTTP load distribution under Statistics tab using Wireshark ver4.4.3.

Evidence 012



Wireshark - HTTP / Load Distribution - XYZ.pcap

Packet Type	Count	Average	Min Val	Max Val	Rate (ms)	Percent	Burst Rate	Burst Start
HTTP Requests by Server	0					100%	-	-
HTTP Requests by HTTP Host	0					-	-	-
HTTP Requests by Server Address	0					-	-	-
HTTP Responses by Server Address	0					100%	-	-

Display filter: `ip.src == 140.247.62.34`

Buttons: Apply, Copy, Save as..., Close

Description: HTTP load distribution pattern with `ip.src == 140.247.62.34` filter on the Statistics tab listing all TCP components detected on the Network. This was obtained on the “HTTP load distribution” under Statistics tab using Wireshark ver4.4.3.

Evidence 013

The image shows the Wireshark 'IPv4 Statistics / Destinations and Ports' window. The left pane shows a tree view with '192.168.15.4' expanded, showing 'TCP' and '140.247.62.34'. The right pane shows a table of statistics for these destinations. The table has columns: Topic / Item, Count, Average, Min Val, Max Val, Rate (ms), Percent, Burst Rate, and Burst Start. The data is as follows:

Topic / Item	Count	Average	Min Val	Max Val	Rate (ms)	Percent	Burst Rate	Burst Start
IPv4 Statistics/Destinations and Ports	117				0.0002	100%	0.1800	13099.258
192.168.15.4	52				0.0001	44.44%	0.0900	13099.258
TCP	52				0.0001	100.00%	0.0900	13099.258
34602	3				0.0000	5.77%	0.0100	13245.897
34600	3				0.0000	5.77%	0.0100	13245.896
34598	3				0.0000	5.77%	0.0100	13238.145
34580	3				0.0000	5.77%	0.0100	12920.821
34560	3				0.0000	5.77%	0.0100	12895.025
34558	3				0.0000	5.77%	0.0100	12895.023
34556	3				0.0000	5.77%	0.0100	12873.687
34554	14				0.0000	26.92%	0.0400	12853.506
34544	3				0.0000	5.77%	0.0100	12827.199
34528	11				0.0000	21.15%	0.0500	13099.258
34526	3				0.0000	5.77%	0.0100	12768.554
140.247.62.34	65				0.0001	55.56%	0.0900	13099.262
TCP	65				0.0001	100.00%	0.0900	13099.262
8000	65				0.0001	100.00%	0.0900	13099.262

Display filter: ip.addr==140.247.62.34

Description: Above image shows the Wireshark's section "Destination and Ports". We can see a complete network traffic analysis here. Here shows the extensive parameters such as packet counts, rates and burst details of highlighted IP addresses 140.247.62.34. So, we can get a clear idea about the traffic flowing through various TCP ports.

6. Conclusions

This detailed network forensic research presents a precise sequence of events that resulted in teacher harassment at the XYZ school. This report reveals vital information about Johnny Coach's activities through the analysis of network traffic data stored in the XYZ.pcap file during a meticulous exam. Network forensic research creates an exact timeline showing how teacher harassment occurred at the XYZ school. The XYZ.pcap file network traffic data analysis under forensic examination provides critical updates about Johnny Coach's activities in this report.

The comprehensive investigation proved that Johnny Coach was the main perpetrator and established the links that tied him to the harassment incident. The comparison of digital evidence against student timetables revealed an exact correlation between Johnny Coach's anonymous online habits and his genuine personal information (True identity).

The research clearly shows that Johnny Coach intentionally disguised his identity through virtual machines and anonymising services.

The forensic analysis conclusions show why digital evidence identification through proper analysis is necessary for investigating cyber bullying cases. Disciplinary and legal responses toward Johnny Coach have become possible since his certain identity as the attacker became clear. Forensic techniques use this article to show their power in describing complex technological events while establishing accountability and protecting against abusive online activities.

7. Self-Review

Minsadhi Vihasna Edirisinghe – s8145500

The report analysis task given to me included examining network tools and IP address 192.168.15.4 activity timeline and handling the received data during the technical investigation. The data processing at my station needed to proceed with strict adherence to established protocols in order to achieve accurate results. It was my responsibility to review background evidence in order to create an understanding of our findings while offering support for the report's final conclusions.

Striving for excellent documentation was my main driving force as I laid out every step process together with all equipment we utilized. Through team collaboration I helped confirm obtained evidence while supporting the synthesis of our research conclusions to produce meaningful results.

The report structure received support from me as I teamed up with my group member Gayathmee to develop a unified presentation for our findings.

Gayathmee Kiveka Nimsandi Balapu Waduge – s8145639

The report analysis task given to me includes, reviewing the network capture file details with network components involved. I identified the key players in the network involved with the situation and the network structure. I carefully followed a structural approach to identify the suspicious student associating the IP address provided with the email header. Finally, I provided clear supporting evidence from network packets.

Through these steps I teamed up with Minsadhi, to provide a clear and evidenced based conclusion regarding the harassment incident.

8. Individual Contribution

Heading	Edirisinghe Minsadhi Vihasna s8145500	Gayathmee Kiveka Nimsandi Balapu Waduge s8145639
Executive summary	100%	0
Network Capture File Details	0	100%
Network Components Identified	0	100%
Tools Used	100%	0
Steps Involved	0	100%
Handling Data	100%	0
Important network players	0	100%
Network Structure	0	100%
Activity Timeline	100%	0
Background Evidence	100%	0
Supporting Evidence	0	100%
Conclusion	100%	0